HACKERSIGNAL: A Large-Scale Multi-Source Dataset Linking Hacker Community Discourse to the CVE Vulnerability Lifecycle

Anonymous Author(s)
Affiliation
Address
email

Abstract

We introduce HackerSignal, a benchmark for temporal out-of-distribution cyber threat intelligence (CTI) and cross-source CVE linkage. HackerSignal aggregates **7.45 million exact-deduplicated documents** from **64 public forum/source identifiers** spanning eight source layers and a 36-year window (1990–2026). In contrast to other publicly accessible cybersecurity datasets, HackerSignal is among the first public benchmark datasets that maps the full potential exploit to vulnerability trajectory from hacker community discourse, exploit databases with working and proof of concept exploits, vulnerability advisories, and software fix commits. HackerSignal creates these linkages through a shared CVE identifier space while preserving source-specific release modes to support a range of unique Artificial Intelligence (AI)-enabled cybersecurity analytics tasks. In this paper, we summarize HackerSignal and illustrate three selected benchmark tasks it uniquely supports: (1) *CVE linkage retrieval* (cross-source temporal out-of-distribution entity grounding); (2) *exploit type classification* (8-class vulnerability type prediction with temporal OOD evaluation); and (3) *temporal generalization* (prospective CVE-disjoint evaluation where $C_{\text{train}} \cap C_{\text{test}} = \emptyset$). All tasks use temporal splits to evaluate prospective generalization. We release source-shortcut and leakage diagnostics, manual-audit packets, a datasheet, and a release-governance addendum to support the dissemination of the dataset. HackerSignal's code, data, and Croissant metadata are available at `hf.co/datasets/BenAmpel/HackerSignal` (data) and `github.com/BenAmpel/hackersignal` (code).

1 Introduction

Understanding how cyber threats emerge, propagate, and relate to vulnerabilities is a central challenge in modern applied cybersecurity. Signals that tracks this trajectory is distributed across heterogeneous text-based sources, including: (1) hacker forums where exploits are first discussed, (2) public exploit databases where proof-of-concept code is released, (3) official vulnerability advisories such as the National Vulnerability Database, CISA's Known Exploited Vulnerabilities, and GitHub Advisory where threats are cataloged, and (4) open-source repositories where fixes are committed. Extant public datasets in this domain often span a single or subset of these layers (Pastrana et al., 2018) and are usually not publicly released (Ampel et al., 2020). A survey of CTI extraction from text identifies advisory records and threat reports as the dominant labeled sources studied, with exploit-archive and PoC-focused benchmarks remaining scarce (Rahman et al., 2023); a companion survey of cybercrime community research corroborates that structured datasets linking forum posts to formal vulnerability records are largely absent from the public literature (Hughes et al., 2024). HackerSignal addresses this gap with a temporally structured, multi-source corpus and benchmark spanning the full

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advisory-to-exploit trajectory, enabling downstream CTI analytics tasks. HackerSignal makes three key contributions:

1. **A temporally grounded CTI corpus** of 7.45M documents from 64 public forum/source identifiers across eight source layers (hacker communities, exploit archives, vulnerability references, exploit Q&A references, advisories, bug-bounty disclosures, fix commits, and exploitation references), with CVE linkage metadata connecting vulnerability identifiers to the texts that discuss them. The 6.97M-row hacker community subcorpus alone (spanning 56 forums across darkweb marketplaces, security discussion boards, and multilingual hacker communities) supports diverse NLP tasks beyond our three benchmarks, including exploit code summarization, security content classification, topic modeling, community interaction analysis, and cross-lingual CTI extraction.
2. **Three benchmark tasks** with temporal train/val/test splits that test progressively harder generalization in (1) cross-source CVE linkage retrieval, (2) 8-class exploit type classification, and (3) CVE-disjoint temporal generalization ($C_{\text{train}} \cap C_{\text{test}} = \emptyset$), accompanied by shortcut/leakage diagnostics.
3. **A full model ladder and governance artifacts** from BM25 through four dense bi-encoders (MiniLM, mpnet, BGE, E5), bag-of-words classifiers (Decision Tree, TF-IDF+LR, SVM), recurrent neural networks (RNN, GRU, LSTM, BiLSTM), three cybersecurity-domain transformers (SecBERT, SecureBERT, CySecBERT), and hybrid sparse+dense retrieval, alongside source-shortcut diagnostics, manual-audit packets, and a release-governance addendum.

Figure 1 provides an overview of the complete pipeline of the proposed dataset, including eight source layers unified through a shared CVE identifier space, then channeled into three benchmark tasks with temporal controls. Temporal splits are necessary in our context as real CTI systems must generalize to threats that did not exist at training time (Koh et al., 2021). We explicitly audit the dataset for temporal and source leakage (Kapoor and Narayanan, 2023) because benchmark claims are otherwise easy to inflate (discussed in Section 3).

2 Related Work

Hacker-community cyber threat intelligence. Early hacker-forum CTI work treated underground communities as a source of proactive intelligence about tools, actors, and emerging attack capabilities (Samtani et al., 2015, 2017). The AZSecure Hacker Assets Portal operationalized this as a situational awareness platform (Samtani et al., 2021), while parallel work modeled threat emergence over time via diachronic graph embeddings (Samtani et al., 2020) and mobile malware identification (Grisham et al., 2017). The resulting artifacts are operational systems rather than reusable ML benchmarks, since their data remains proprietary. Rahman et al. (2023) survey automated CTI extraction from text, observing that collections often concentrate on the advisory and threat-report tier, while exploit-archive and PoC-focused benchmarks remain scarce. Hughes et al. (2024) systematize research on the cybercrime community and demonstrate a persistent shortage of ground-truth datasets. Most publicly collected forum data lacks structured labels linking posts to formal vulnerability records.

Exploit labeling and vulnerability linkage. The closest predecessor to HackerSignal is in the field of exploit labeling research. Prior methods labeled hacker-forum exploit source code via deep transfer learning from professional exploit repositories (Ampel et al., 2020) and later extended this with pre-initialization, multi-layer transfer, and self-attention (Ampel et al., 2024). For cross-source linkage, an attention-based deep-structured semantic model linked dark-web exploits to known vulnerabilities, yielding 52,590 linkages (Samtani et al., 2022). HackerSignal differs by releasing a governed, temporally split public benchmark that exposes source provenance, weak-label caveats, and a quality-controlled CVE-linkage retrieval task. Finally, the hacker forum collection of Otto et al. (2021) aggregates 259K posts from three English-language security communities (AntiOnline, Go4Expert, CipherSpace) and studies exploit-sharing behavior via graph embeddings.

Underground community datasets. Research has also focused on the general crawling of hacker communities, like a credentialized dataset of darkweb cybercrime forum posts widely used for

crime and exploit prediction (Pastrana et al., 2018) Nunes et al. (2016) presented an operational darknet/deepnet mining system over hacker forums and markets for proactive CTI. The Gayanku (Gayanku, 2020) dataset contains 3.8M posts from seven darknet marketplaces (Silk Road, Agora, Evolution, etc.) on Kaggle.

Exploit and vulnerability databases. ExploitDB (Offensive Security, 2003–2026) is the salient public exploit archive (50K+ entries, many with CVE IDs). The CVEfixes dataset (Bhandari et al., 2021) links CVEs to fix commits in open-source repositories. NVD (NIST, 2002–2026), CISA KEV (CISA, 2021–2026), and GitHub Advisory (GitHub, 2017–2026) are official advisory sources. HackerSignal normalizes all five into a common JSONL schema with CVE keys.

Multilingual CTI. More general hacker-forum text classification work compared SVM and CNN models for extracting CTI from forum posts (Deliu et al., 2017), while Ebrahimi et al. (2020) showed that non-English hacker forums require cross-lingual approaches that avoid brittle machine translation. CVE severity prediction (Han et al., 2017), exploit labeling (Ampel et al., 2024), and exploit-vulnerability linkage (Samtani et al., 2022) all motivate our task design; HackerSignal operationalizes them as public, temporally split benchmarks.

Benchmark design and evaluation method. BEIR (Thakur et al., 2021) established that retrieval benchmarks require heterogeneous baseline ladders rather than single-model comparisons. KILT (Petroni et al., 2021) showed that knowledge-intensive linking tasks should reward both task success and provenance quality. WILDS (Koh et al., 2021) formalized natural arising distribution shifts (including temporal shifts) as first-class benchmark objects. At the same time, WRENCH (Zhang et al., 2021) demonstrated that weak-supervision evaluation is sensitive to hidden protocol choices, requiring explicit reporting of label sources and aggregation rules. Kapoor and Narayanan (2023) documented widespread data leakage in ML-for-science benchmarks and proposed structured reporting to make integrity checks legible. These methodological precedents directly inform our temporal splits, source-shortcut diagnostics, and audit artifacts.

3 Dataset

3.1 Source Layers

HackerSignal integrates public hacker-community discourse with sources of exploit, advisory, vulnerability-reference, bug-bounty, and commit-fix in eight source layers (Figure 1, left). Table 1 reports the composition of the audited release; Table 2 reports quality diagnostics.

Hacker and security communities (6.97M rows). The hacker-community layer captures organic security discourse. This includes, but is not limited to, discussions of exploits, tool sharing, vulnerability commentary, marketplace posts, and peer support across communities. Our collection combines historical forum datasets (e.g., Gayanku, Evolution, and HackForums) with newer publicly accessible sources such as DeepDarkCTI communities, 0x00sec, HackerSploit, and ParrotSec. The largest forums after deduplication are `gayanku_darkweb` (3.11M rows), `antichat` (1.82M), `evolution_darkweb` (471K), `antionline` (463K), `persiantools` (280K), `wwhclub` (161K), and `hackforums` (151K). Forum sources often carry sparse direct CVE mentions and exhibit heterogeneous vocabulary overlap across communities (Figure 3). These characteristics make CVE linkage from community discourse a nontrivial retrieval problem.

Reference and supervision layers (482K rows). The remaining layers provide CVE-grounded supervision and comparison corpora. NVD CVE descriptions (340,536 rows) form the retrieval corpus for all three benchmark tasks. ExploitDB, 0day.today, Packet Storm, ZeroScience, and Vulnerability-Lab contribute exploit and advisory text. GitHub Advisory, CISA KEV, HackerOne, and CVEfixes contribute structured records for vulnerabilities, bug bounties, known-exploited issues, and fix commits. These sources are separated by `source_layer`; thus, experiments can evaluate either community-only discourse or community-plus-context settings.

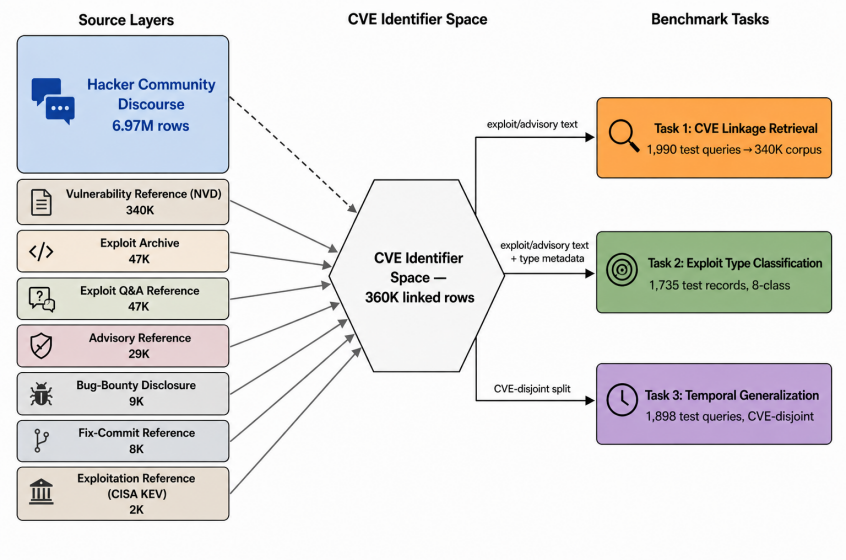


Figure 1: HackerSignal pipeline overview. Dashed arrows indicate sparse, indirect CVE mentions. Solid arrows indicate direct CVE metadata linkage.

Table 1: Dataset statistics after exact deduplication. The hacker community layer is expanded by forum. The remaining layers are single-source. Avg. Len = mean whitespace tokens; CVE% = rows containing an explicit CVE identifier.

Source Layer	Source / Forum	#Docs	Avg Len	CVE%	Span
Hacker Community	gayanku_darkweb	3,107,038	72.6	<0.1	2011–17
	antichat	1,819,730	39.7	<0.1	2002–26
	evolution_darkweb	471,115	65.4	<0.1	2014–15
	antionline	462,510	73.6	<0.1	2001–26
	persiantools	280,293	82.1	<0.1	2026
	wwhclub	161,026	51.3	<0.1	2026
	hackforums	151,072	31.6	<0.1	2009–15
	go4expert	110,345	75.1	<0.1	2004–26
	Other (48 forums)	402,885	56.5	0.1	2000–26
	Subtotal	**6,966,014**	**61.7**	**<0.1**	**2001–26**
Vulnerability Ref (NVD)		340,536	47.2	100.0	2002–26
Exploit Archive		46,653	240.0	10.5	1991–26
Exploit Q&A Ref		46,505	25.2	<0.1	2000
Advisory Ref		28,684	110.8	5.0	2017–26
Bug-Bounty Disclosure		9,453	191.1	6.5	2013–24
Fix-Commit Ref		8,293	92.0	100.0	1999–22
Exploitation Ref (KEV)		1,508	54.4	3.8	2021–26
Total (64 sources)		**7,447,646**	**62.0**	**4.8**	**1991–26**

3.2 CVE Identifier Space

All sources are unified through a common JSONL record format and a shared CVE identifier namespace (Figure 1, center). Each record carries a SHA-256-derived release identifier, source provenance (`source_layer`, `forum_id`), cleaned UTF-8 text, an ISO 8601 timestamp, and a pseudonymized author hash. A separate CVE index maps each record to its CVE reference(s) via a foreign key, enabling cross-source joins through the shared identifier space. Of the 7.45M documents, 360,004 rows (4.83%) contain explicit CVE references, forming the linkage substrate for all three benchmark tasks. The full schema specification is provided in the accompanying datasheet (Gebru et al., 2021).

Table 2: Corpus quality diagnostics.

Diagnostic	Rows	Share
Parseable timestamps	7,447,646	100.0%
CTI-keyword nonzero	1,748,049	23.5%
CVE string present	360,004	4.8%
Short rows (<8 tok)	1,569,263	21.1%
Long rows (>3000 tok)	815	<0.1%
Est. near-duplicates	609,029	8.2%

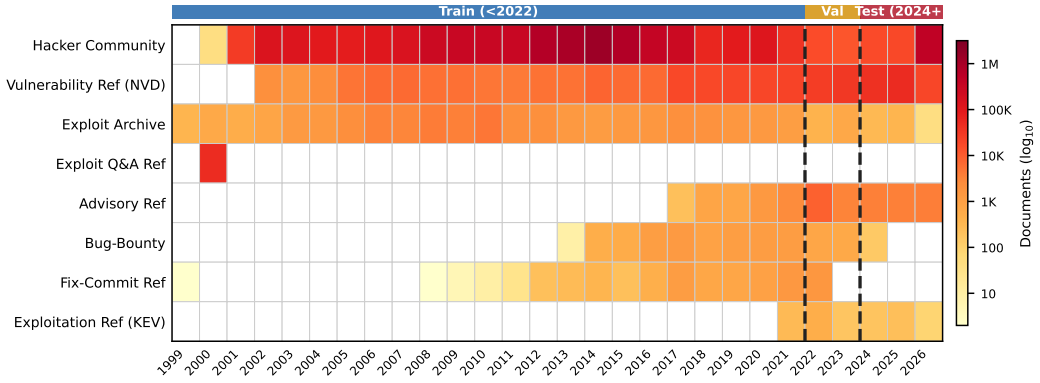


Figure 2: Temporal coverage by source layer ($\log_{10}$ scale). Colored bars indicate benchmark split boundaries: train (<2022), val (2022–23), test (2024+).

3.3 Collection and Preprocessing

Data was collected between January and April 2026 via (i) official APIs (NVD, GitHub Advisory, CISA KEV, HackerOne HuggingFace Hub), (ii) ingestion of existing public datasets (e.g., Gayanku, HackForums, CVEfixes, DeepDarkCTI, CrackingArena), and (iii) ethical web scraping of publicly indexed sites (e.g., ExploitDB, Vulnerability-Lab, ZeroScience, 0x00sec, HackerSploit, ParrotSec, Hack The Box, Go4Expert, Full Disclosure). All collection code is released alongside the dataset.

We took four steps to prepare the data for our three proposed tasks. First, we perform text cleaning by normalizing Unicode, stripping HTML tags, replacing URLs, and collapsing extra whitespace. Second, we filtered out low-quality content, including non-English language and posts shorter than 20 characters. Third, we run exact deduplication via SHA-256 hashing and near-deduplication via MinHash-LSH with 128 permutations, 3-character k-shingles, and Jaccard similarities between posts over 0.7. Finally, we divided the posts into temporal train, validation, and test splits. The release removes 814,922 posts. We show the final dataset distribution in Figure 2.

Several high-volume historical sources (HackForums, Gayanku, Evolution, and the AntiOnline/Go4Expert forums) end in 2015-2017, while newer sources (DeepDarkCTI communities, 0x00sec, HackerSploit, ParrotSec, NVD, GitHub Advisory, and CISA KEV) extend into 2026 (Figure 2). In addition, the corpus is predominantly English. Multilingual content (Russian, Turkish, Persian, Arabic, Chinese) appears in DeepDarkCTI and several dark-market sources. Short rows (<8 tokens, 21.1%) include replies and marketplace acknowledgments. These should be filtered for classification and retrieval tasks, but are useful for other analyses of community interactions.

Past research indicates that hacker community platforms vary in the content they offer, such as exploits and tools. To measure lexical similarity among 46 forums, we compute pairwise Jaccard coefficients using a 5% token sample and perform hierarchical clustering (Figure 3). The heatmap and network show a dense cluster of English-language security forums (hackforums, zerosec, cipher, wilderssecurity, go4expert, cipherspace) with shared technical vocabulary. Dark Web marketplaces (Gayanku, Evolution) form a second cluster connected by transactional terms. Non-English forums (antichat, persiantools) and niche groups (parrotsec, zeroday today) are peripheral with low overlap ($J < 0.06$), showing language and domain create vocabulary boundaries. Node sizes reflect contribution on a log_{10} scale, highlighting antichat, gayanku, and wwhclub's prominence in document

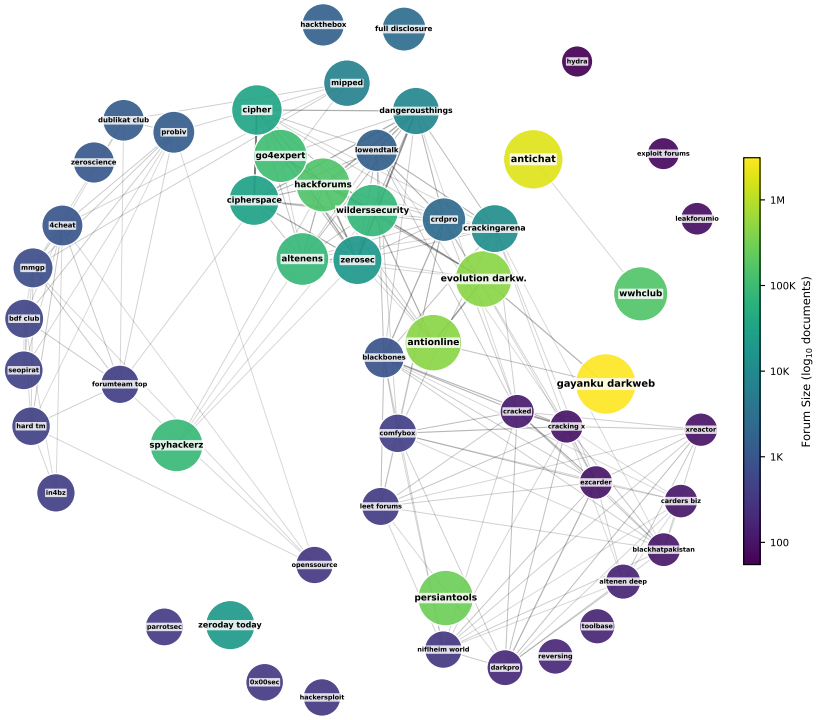


Figure 3: Pairwise vocabulary overlap across 46 hacker community forums (5% deterministic token sample, hierarchically clustered).

Table 3: Benchmark task summary. Task 3 splits by CVE publication year (train <2022, val 2022–2023, test 2024+) and Task 1 uses the same test queries by source date (92 of its 1,990 test queries link to CVEs published before 2024), both over the full 340K NVD corpus; Task 2 uses an earlier split (train <2020, test ≥2023).

Task	Type	Train	Val	Test	Corpus / Total
1. CVE-R	Retrieval	56,692	2,584	1,990	340,536
2. ETC	8-class classif.	64,413	4,735	1,735	70,883
3. TG	CVE-disjoint Retr.	56,833	2,535	1,898	340,536[†]

[†]Same corpus; zero CVE overlap between Task 3 train/val/test gold sets.

count despite lexical isolation. This heterogeneity supports our evaluation, as models must generalize across diverse vocabularies.

4 Benchmark Tasks and Results

HackerSignal provides a unique longitudinal, multi-lingual, and multi-sourced CTI dataset that can enable novel research opportunities examining temporal phenomena, generalization, and more. We illustrate three sample CTI tasks with increasing levels of methodological complexity that HackerSignal enables in (Figure 1, right; Table 3). Each probes a different facet of temporal out-of-distribution generalization. All tasks use temporal splits, ensuring that test data postdate the training data. Following Thakur et al. (2021), we evaluate each task with a heterogeneous baseline ladder spanning lexical, dense, and hybrid methods.

All benchmark labels are derived from platform-provided and standardized metadata rather than manual annotation (helping reduce third-party biases). Following Zhang et al. (2021), we explicitly report weak-label sources and aggregation rules, as minor protocol changes can alter benchmark results and conclusions. For retrieval tasks (Tasks 1 and 3), a query's gold CVE is the metadata-linked

entry from its source. This may be incomplete or linked to a related rather than maximally specific CVE. For exploit type classification (Task 2), labels are derived from structured metadata fields (ExploitDB exploit type and description, exploit type, HackerOne vulnerability type prefix) and mapped to a unified 8-class taxonomy via deterministic rules. We provide a 240-row stratified manual audit packet in `data/audits/` for transparency.

4.1 Task 1: CVE Linkage Retrieval (CVE-R)

Let $\mathcal{Q}$ be a set of exploit/advisory evidence texts and $\mathcal{C} = \{c_1, \ldots, c_N\}$ the NVD corpus of N=340,536 CVE descriptions. For each query $q \in \mathcal{Q}$, the task is to retrieve the correct CVE entry $c^* \in \mathcal{C}$ such that the source metadata links q to c^*. This is cross-source temporal out-of-domain entity grounding, akin to knowledge-intensive entity linking (Petroni et al., 2021), except the target knowledge base is the CVE ecosystem. Queries come from exploit archives, advisories, fix commits, and bug bounty reports, while the target space is a single, canonical CVE registry. We retain queries from 22 source identifiers spanning exploit archives, advisories, fix commits, and bug bounty reports (see `data/benchmark_v2/task1_cve_linkage/meta.json` for the full list) and require $\geq$8 query tokens. Labels are metadata-derived linkages, not manually adjudicated ground truth.

We choose baselines across three dense retrieval paradigms. BM25 is an unsupervised lexical baseline that serves as an exact-match ceiling. For dense bi-encoders, we selected four models differing in pre-training and architecture: (1) MiniLM-L6-v2 (speed-optimized, distilled 6-layer), (2) all-mpnet-base-v2 (mean-pooled MPNet with contrastive training on 1B+ pairs), (3) BGE-base-v1.5 (instruction-tuned bi-encoder from BAAI for zero-shot transfer), and (4) E5-base-v2 (weakly-supervised contrastive model with task-specific prompts). These span distilled vs. full-size and supervised vs. weakly supervised training, helping assess the impact of pre-training on CVE retrieval. Hybrid BM25+mpnet reranks the BM25 top-100 candidates with a weighted sum of normalized BM25 and dense cosine scores (α=0.7 on the dense score), testing their complementarity. We measure Recall@k ($k = 10$) and MRR, with results in Table 4.

Table 4: Task 1 (CVE-R) test results (1,990 queries, 340K corpus). Bootstrap 95% intervals on MRR@10 (10,000 resamples) have a half-width of about 0.02; the top two MRR values do not differ significantly.

Model	R@1	R@5	R@10	MRR
BM25	0.595	0.688	0.715	0.636
MiniLM-L6-v2	0.655	0.776	0.809	0.709
mpnet-base-v2	0.626	0.756	0.792	0.682
BGE-base-v1.5	0.653	**0.789**	0.823	**0.714**
E5-base-v2	0.649	**0.789**	**0.824**	0.710
Hybrid BM25+mpnet	**0.679**	0.753	0.768	0.711

All dense models and the hybrid exceed BM25 (MRR 0.636) and cluster tightly (MRR 0.682–0.714). BGE-base-v1.5 has the highest MRR (0.714), but its difference from the hybrid (0.711) is not significant (paired ΔMRR 0.003, 95% CI [$-$0.013, 0.019]). No single model dominates: the hybrid leads at R@1 (0.679), and BGE and E5 lead at R@5 and R@10.

4.2 Task 2: Exploit Type Classification (ETC)

Given an exploit text x, predict its vulnerability type $y \in \mathcal{Y}$ where $|\mathcal{Y}| = 8$ classes: *injection* (SQLi, command/code injection, XXE), *XSS* (reflected, stored, DOM), *memory corruption* (buffer/heap overflow, use-after-free, format string), *denial of service*, *file inclusion* (LFI/RFI, path traversal, arbitrary file operations), *authentication & access control* (privilege escalation, auth bypass, CSRF, IDOR), *remote code execution* (RCE, deserialization), and *information disclosure* (SSRF, open redirect, data exposure). This taxonomy aligns with CWE top-level categories and prior exploit classification work (Ampel et al., 2020, 2024). Labels are derived from structured metadata: ExploitDB exploit descriptions and tags, 0day.today exploit type fields, and HackerOne vulnerability type prefixes. The dataset comprises 70,883 labeled exploit texts from these three source families.

We use a temporal out-of-distribution split (train $<$2020, val 2020-2022, test $\geq$2023) so that models must generalize to vulnerabilities and exploits that postdate training. We evaluate a model ladder spanning bag-of-words classifiers (Decision Tree, TF-IDF+LR, linear SVM), recurrent neural networks (RNN, GRU, LSTM, BiLSTM), and three cybersecurity-domain transformers (SecBERT (Jackaduma, 2022), SecureBERT (Aghaei et al., 2023), and CySecBERT (Bayer et al., 2024)). We report macro-F1 (mean $\pm$ SD over three seeds) and per-class F1 for the three hardest classes in Table 5.

Table 5: Task 2 (ETC) test results on 1,735 temporally held-out samples. Macro-F1 is the mean ± SD over three seeds; per-class F1 for the three hardest classes is the seed mean. Bootstrap 95% intervals on macro-F1 (2,000 resamples) have a half-width of about 0.02; the top two models do not differ significantly. BoW = bag-of-words features.

	Model	Macro-F1	info_disc.	mem_corr.	auth.
BoW	Decision Tree	.702 ± .007	.45	.64	.60
	TF-IDF + LR	.772 ± .000	.66	.69	.79
	SVM (Linear)	.799 ± .000	.70	.68	.82
Recurrent	RNN	.436 ± .031	.30	.42	.44
	GRU	.858 ± .001	.70	.84	.81
	LSTM	.824 ± .013	.63	.77	.78
	BiLSTM	.856 ± .003	.69	.84	.81
Transf.	SecBERT	.852 ± .007	.67	.82	.83
	SecureBERT	.892 ± .008	**.80**	.90	.85
	CySecBERT	**.898 ± .003**	.79	**.90**	**.86**

The best bag-of-words baseline (SVM) attains macro-F1 of 0.799. Gated recurrent models improve on it: GRU (0.858) and BiLSTM (0.856) are level, and both exceed LSTM (0.824; GRU−LSTM paired Δ 0.029, 95% CI [0.016, 0.043]). The vanilla RNN is far weaker (0.436): 32% of test exploits exceed the 300-token window, and without gating the signal from early tokens fades over long sequences. Domain pre-training helps most. CySecBERT (0.898) and SecureBERT (0.892) are the strongest models, and their difference is not significant (paired Δ 0.006, 95% CI [−0.004, 0.016]), while SecBERT (0.852) is level with the best recurrent models. The hard classes (information disclosure, memory corruption, authentication/access) have substantial overlap. For example, disclosure exploits often involve injection (SSRF, XXE), while privilege escalation and authentication share patterns with RCE. Decision Tree drops to macro-F1=0.702, with disclosure at 0.45. Lexical classes such as XSS and injection are easier (mean F1 0.94 and 0.86 across models), indicating that exploit signals are partly lexical. Temporal splits make it harder as exploit names and vulnerability patterns evolve, preventing models from memorizing era-specific features.

4.3 Task 3: Temporal Generalization (TG)

Identical retrieval formulation to Task 1, but with a strict CVE-disjointness constraint: $C_{\text{train}} \cap C_{\text{test}} = \emptyset$, where $C_s = \{c_i^* : q_i \in \text{split } s\}$ is the set of gold CVE targets in split s. Splits are defined by CVE publication year rather than query timestamp: train uses CVEs published before 2022, validation uses CVEs published in 2022–2023, and test uses CVEs published in 2024+. This ensures that models cannot memorize vulnerability descriptions seen during training and must generalize to wholly unseen vulnerabilities. We use the same metrics and models as Task 1 and show the results of Task 3 in Table 6.

Under the CVE-disjoint evaluation, the hybrid has the highest MRR (0.730), but its difference from BGE-base-v1.5 (0.726) is not significant (paired ΔMRR 0.004, 95% CI [−0.012, 0.020]). Every model scores slightly higher than on Task 1 (e.g., BM25 MRR 0.660 vs. 0.636). The Task 3 test queries are a subset of the Task 1 test queries, so the two tasks differ in the constraint they place on training rather than in the difficulty of the evaluation.

Table 6: Task 3 (TG) results (1,898 test queries) under strict CVE disjointness ($C_{\text{train}} \cap C_{\text{test}} = \emptyset$). Bootstrap 95% MRR intervals have a half-width of about 0.02; the top two MRR values do not differ significantly.

Model	R@1	R@5	R@10	MRR
BM25	0.620	0.713	0.740	0.660
MiniLM-L6-v2	0.670	0.788	0.820	0.722
mpnet-base-v2	0.647	0.777	0.809	0.703
BGE-base-v1.5	0.667	0.798	0.828	0.726
E5-base-v2	0.663	**0.799**	**0.830**	0.721
Hybrid BM25+mpnet	**0.698**	0.772	0.786	**0.730**

4.4 Audit Artifacts and Error Analysis

Following Kapoor and Narayanan (2023), we explicitly audit for temporal and source leakage because benchmark claims are otherwise too easy to inflate. The CVE-disjointness of

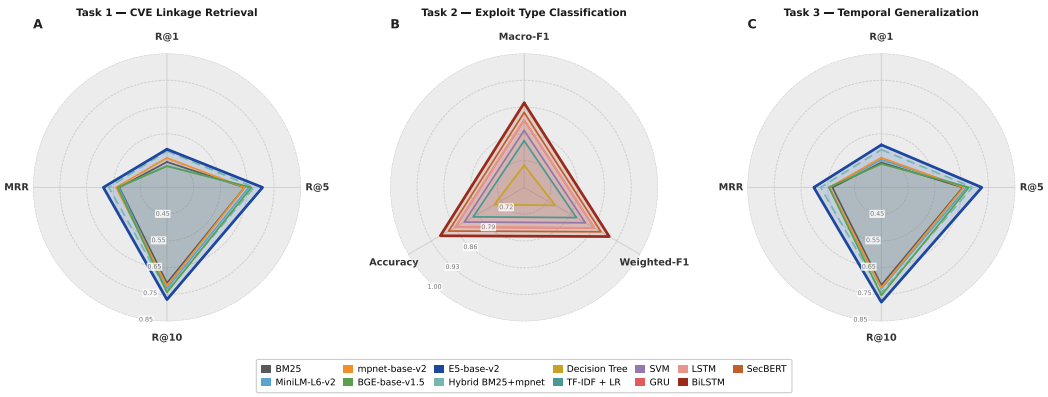


Figure 4: Radar chart summary of baseline performance across all three tasks.

Task 3 is verified programmatically: $|C_{\text{train}} \cap C_{\text{test}}| = 0$. Full details are in Appendix B.

Among the Hybrid model's Task 1 errors (queries where the target CVE does not appear in the top-10), we identify three recurring failure patterns: (1) *Generic exploit descriptions* matching multiple CVEs sharing a common vulnerability class; (2) *Version-specific queries* referencing product versions absent from the NVD description; and (3) *Alias mismatches* where queries use informal names (e.g., Log4Shell) while the NVD entry uses only formal identifiers.

5 Ethical Considerations

The source mix contains various licenses and terms of service. The public release categorizes sources into redistributable, research-only, metadata/pointer, and excluded. A governance addendum details release modes, contact info, versioning, checksums, and takedown procedures. All sources are accessible without authentication, but privacy risks remain. We pseudonymize author IDs via SHA-256, preserve provenance, and support takedown requests, downgrading ambiguous sources accordingly. HackerSignal highlights vulnerabilities for defensive research only, prohibiting fine-tuning automated exploit models, de-anonymizing forum users, and automated incident response without human review. The governance addendum enforces these rules, including release modes and version management. Researchers should consult IRBs when analyzing individuals. The dataset, available on HuggingFace, is for academic and defensive use only, not operational threat feeds. Outputs should not be trusted as the current CTI without validation. The repository includes metadata, a dataset card, and a link to the governance addendum.

6 Conclusion

HackerSignal links 7.45M documents from 64 sources across the CVE lifecycle with temporal splits, CVE-disjoint evaluation, and provenance-aware governance. Our three-task benchmark and model ladder (BM25 through dense and hybrid retrieval at MRR 0.71, domain-pretrained transformers at macro-F1 0.90 for 8-class exploit type classification) establishes baselines for cross-source CVE grounding, exploit type classification, and prospective generalization to unseen vulnerabilities.

Several limitations bound the current release. First, CVE linkages are metadata-derived (source URLs, CVE regex extraction) rather than expert-adjudicated, introducing noise proportional to source quality. We mitigate this via per-source audit risk scores, but cannot guarantee zero false linkages. Second, although some hacker forums contain non-English posts, we do not evaluate cross-lingual retrieval or multilingual baselines. Prior work suggests non-English forums require dedicated approaches (Ebrahimi et al., 2020). Third, we do not evaluate large language model rerankers (e.g., GPT-4, Claude) or retrieval-augmented generation pipelines, which may score higher but at orders-of-magnitude higher compute cost.

Beyond the three tasks benchmarked here, the 6.97M-row hacker community subcorpus opens research directions in exploit code summarization and completion, hacker content extraction and labeling, topic modeling (e.g., LDA over forum-specific vocabularies), community interaction network analysis, cross-lingual CTI transfer across the corpus's multilingual forums, and temporal trend detection in exploit discourse. HackerSignal's code, data, splits, and Croissant metadata are publicly released to support these extensions at `hf.co/datasets/BenAmpel/HackerSignal` (data) and `github.com/BenAmpel/hackersignal` (code).

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Table 7: Per-source collection details.

Source	Method	Access	License	Release mode
NVD	REST API	Public	Public domain	Text
CISA KEV	JSON feed	Public	Public domain	Text
GitHub Advisory	GraphQL API	Public	CC BY 4.0	Text
ExploitDB	CSV download	Public	CC BY-SA 4.0	Text
HackerOne	HuggingFace Hub	Public	Source dataset terms	Research text
CVEfixes	Zenodo SQLite	Public	CC BY 4.0	Text
Kaggle forums	Kaggle	Public	Dataset-specific	Research/pointer
DeepDarkCTI	GitHub	Public	Repository terms	Research text
0x00sec/HTB/etc.	Discourse API	Public	Site terms	Pointer
Other web archives	Web scraping	Public	Mixed terms	Pointer

A Source Collection Details

The full source-by-source matrix and takedown/correction process are in `docs/release_governance.md`. Collection scripts are available at `github.com/BenAmpel/hackersignal` under `src/etg/collectors/`. The public JSONL release is built from the full research build by enforcing `release_mode`: rows from `metadata_or_pointer_only` sources retain provenance, timestamps, source/thread identifiers, CVE references, text hashes, and text lengths, but `text` and `text_raw` are withheld. Table 7 provides per-source collection method, access mechanism, license posture, and default release mode.

B Audit and Reliability

We follow the audit framework of Kapoor and Narayanan (2023) and perform three complementary integrity checks. First, for each task, we draw an 80-row stratified random sample (240 rows total, balanced across splits) and provide a structured annotation packet in `data/audits/` covering (i) correct CVE assignment, (ii) temporal split adherence, and (iii) absence of trivial textual overlap between query and target. Programmatic checks on the sampling frame found no structural errors. Second, we search for exact and near-duplicate text spans between train and test partitions using MinHash (Jaccard threshold $\geq$0.8, 128 permutations). After deduplication during corpus construction, zero train-test pairs exceed the threshold for any task. Finally, we check gold-CVE overlap between splits. For Task 3 (temporal generalization), CVE-disjointness is enforced by construction: $|C_{\text{train}} \cap C_{\text{test}}| = 0$, supported programmatically at build time. Task 1 does not impose disjointness; 9 of its 1,876 test CVEs (9 of 1,990 test queries) also occur in training.

C Model Ladder Details

The full model ladder evaluates six retrieval approaches (Tasks 1 and 3) and ten classification approaches (Task 2). Each model represents a rung on the complexity ladder from lexical to neural to hybrid methods. BM25 uses no learned parameters; bi-encoders (MiniLM, mpnet, BGE, E5) use pre-trained sentence embeddings without task-specific fine-tuning; the hybrid combines BM25 top-100 reranking with dense scores (α=0.7). For Task 2, the ladder spans bag-of-words classifiers (Decision Tree, TF-IDF + LR, linear SVM), recurrent neural networks (RNN, GRU, LSTM, BiLSTM), and three domain-specific transformers (SecBERT, SecureBERT, CySecBERT).

D Baseline Hyperparameter Details

BM25 (Tasks 1 and 3). Okapi BM25 via `rank_bm25` (BM25Okapi). Tokenization: lowercase whitespace split. Full 340,536-record corpus indexed.

Dense bi-encoders (Tasks 1 and 3). Four sentence-transformer models evaluated without task-specific fine-tuning: `all-MiniLM-L6-v2` (22M params, 384-dim), `all-mpnet-base-v2` (110M, 768-dim), `BAAI/bge-base-en-v1.5` (110M, 768-dim), and `intfloat/e5-base-v2` (110M, 768-dim, with "query:"/"passage:" prefixes). Inputs are the first 384 characters of each query and

document, max sequence length 512 tokens, batch size 128, ℓ_2-normalized fp32 embeddings, cosine similarity scoring; BGE is used without its query instruction. MRR is computed over the top 10 (MRR@10), with ties broken by corpus order. Two settings matter: encoding full texts instead of the first 384 characters raises Task 1 MRR for the base-size models by about 0.03 (e.g., BGE 0.744, E5 0.741) and lowers MiniLM by 0.025, and adding BGE's query instruction raises its MRR to 0.725 (Task 1) and 0.737 (Task 3). E5's prefixes change MRR by less than 0.001.

Hybrid BM25 + dense (Tasks 1 and 3). A combination of normalized BM25 scores and dense cosine similarity with α=0.7 (dense weight). For efficiency, BM25 retrieves the top 100 candidates per query; dense scores are computed only for the candidate set. Final ranking uses $\alpha \cdot \text{dense} + (1-\alpha) \cdot \text{BM25}$.

Bag-of-words classifiers (Task 2). All BoW classifiers use scikit-learn with `TfidfVectorizer`(max_features=50,000, ngram_range=(1,2)) as the feature extractor, applied to the first 2,000 characters of each text. `DecisionTreeClassifier`: max_depth=30, class_weight="balanced". `LogisticRegression`: C=1.0, max_iter=1000, class_weight="balanced". `LinearSVC`: max_iter=2000, class_weight="balanced".

Neural classifiers (Task 2). RNN, GRU, LSTM, and BiLSTM use a word-level vocabulary (min frequency $\geq$3, yielding 175,062 tokens) with 128-dim embeddings, 128-dim hidden states, max sequence length 300 (right-padded), batch size 128, and Adam optimizer (lr=10^{-3}). The classifier reads the hidden state at the last non-padding token (both directions for BiLSTM, via packed sequences). Class-weighted cross-entropy loss with gradient clipping at 1.0. Trained for 8 epochs with best checkpoint selected by validation macro-F1. SecBERT (Jackaduma, 2022) (`jackaduma/SecBERT`), SecureBERT (Aghaei et al., 2023) (`ehsanaghaei/SecureBERT`), and CySecBERT (Bayer et al., 2024) (`markusbayer/CySecBERT`) are fine-tuned on the first 1,000 characters of each text with max sequence length 256, batch size 16, AdamW (lr=2×10^{-5}, weight decay 0.01), linear warmup over 10% of steps, and unweighted cross-entropy for 3 epochs on a 15K training subsample; the epoch is selected by validation macro-F1. All Task 2 models are run with seeds 42, 43, and 44 (the 15K subsample is drawn per seed); we report the mean and SD. Fine-tuning on all 64,413 training rows is reported in the released results.

Compute. All baselines ran on one Apple M5 Pro workstation (48 GB unified memory, macOS 27) with Python 3.12.15, PyTorch 2.14.1 (MPS backend), sentence-transformers 6.2.0, transformers 5.21.0, and scikit-learn 1.9.1; no CUDA GPUs were used. BM25 uses an exact sparse reimplementation of `rank_bm25` (maximum score difference 3×10^{-12} on a 50-query check): 5–8 s to index the 340K corpus and 15–25 s to score each test set. Dense encoding of the 340K corpus took $\sim$4 min (MiniLM) to $\sim$30 min (base-sized models). Task 2 training per seed: BoW 7–40 s; RNN/GRU/LSTM 5–36 min; BiLSTM $\sim$60 min; SecBERT $\sim$17 min and SecureBERT/CySecBERT $\sim$40–45 min on the 15K subsample. Per-run timings, seeds, model revisions, and data hashes are in `results/camera_ready/provenance.json`. All baseline code is available at `github.com/BenAmpel/hackersignal`.

E Comparison with Prior CTI Datasets

Table 8 positions HackerSignal against prior cyber threat intelligence datasets. We emphasize four differentiators: (i) governed public release with source-specific access modes; (ii) temporal benchmark splits enabling prospective evaluation; (iii) cross-source CVE linkage connecting hacker discourse to formal vulnerability records; and (iv) scale exceeding prior resources by an order of magnitude.

F Datasheet

A complete datasheet is provided in the supplemental material and in the dataset repository at `docs/datasheet.md` (Gebru et al., 2021).

Table 8: Comparison of HackerSignal with prior CTI datasets across key dimensions. ✓ = yes, ✗ = no, ∼ = partial.

Dataset	Size	Sources	Public	CVE Link	Temp. Splits	Governance	Benchmark
CrimeBB (Pastrana et al., 2018)	48M posts	4 forums	✗	✗	✗	✗	✗
Gayanku (Gayanku, 2020)	3.8M posts	7 markets	✓	✗	✗	✗	✗
Otto et al. (Otto et al., 2021)	259K posts	3 forums	✓	✗	✗	✗	✗
ExploitDB (Offensive Security, 2003–2026)	50K+	1 archive	✓	∼	✗	✓	✗
CVEfixes (Bhandari et al., 2021)	8.3K	commits	✓	✓	✗	✓	✗
HackerSignal (ours)	**7.45M**	**64 sources**	✓	✓	✓	✓	✓

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states three concrete contributions (corpus, benchmark tasks, baselines) that are fully realized in Sections 3 and 4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 3 discusses data quality (duplicates, temporal imbalance, label noise) and Section 6 enumerates four explicit limitations: label provenance, English dominance, missing LLM baselines, and compute constraints.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [N/A]

   Justification: This is a dataset paper with no theoretical results.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: All baseline hyperparameters are reported in Appendix D. Code is released at `github.com/BenAmpel/hackersignal`. Data is publicly available on HuggingFace Hub.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code?

   Answer: [Yes]

   Justification: Dataset released on HuggingFace with provenance metadata, Croissant metadata, and a release-governance addendum. Collection and baseline code released at `github.com/BenAmpel/hackersignal`. Accessible without personal request to the PI, with pointer-only modes for sources whose terms do not support text mirroring.

6. **Experimental setting/details**

   Question: Does the paper specify all training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Split boundaries, sources, and hyperparameters are specified in Section 4 and Appendix D.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or statistical significance?

Answer: [Yes]

Justification: Tables 4–6 report test-set bootstrap 95% intervals (10,000 resamples for MRR, 2,000 for macro-F1) and paired bootstrap tests for the top two models; Task 2 reports the mean $\pm$ SD over three seeds. BM25 and dense-retrieval baselines are deterministic given a fixed corpus and queries (ties are broken by corpus order). MPS training is not bit-deterministic, which the seed SD captures.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [Yes]

   Justification: Baselines were run on one Apple M5 Pro workstation (48 GB) with PyTorch 2.14.1 on the MPS backend. Appendix D reports wall-clock times per model and task.

9. **Code of ethics**

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    Question: Does the paper describe safeguards for responsible release?

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12. **Licenses for existing assets**

    Question: Are existing assets properly credited with licenses?

    Answer: [Yes]

    Justification: Appendix A lists the license for every source. The References section cites all dataset papers.

13. **New assets**

    Question: Are new assets well documented?

    Answer: [Yes]

    Justification: A Gebru et al. datasheet is provided (Appendix F and `docs/datasheet.md`). Croissant machine-readable metadata is distributed with the HuggingFace repository.

14. **Crowdsourcing and research with human subjects**

    Question: Does the paper include details for crowdsourcing or research with human subjects?

    Answer: [N/A]

    Justification: No crowdsourcing was conducted. The release uses public OSINT sources, pseudonymizes authors, supports takedown/correction requests, and instructs researchers making individual-level inferences to seek their own IRB review.

15. **Institutional review board (IRB) approvals**

    Question: Does the paper describe IRB approvals for research with human subjects?

    Answer: [N/A]

Justification: Collection of public OSINT data was not classified as human-subjects research under institutional guidelines; the paper still documents privacy safeguards, responsible-use restrictions, and release-mode controls.

16. **Declaration of LLM usage**

Question: Does the paper describe usage of LLMs as an important component of the core methods?

Answer: [Yes]

Justification: LLMs were not used to create benchmark labels. Writing assistance was also used for editing.